

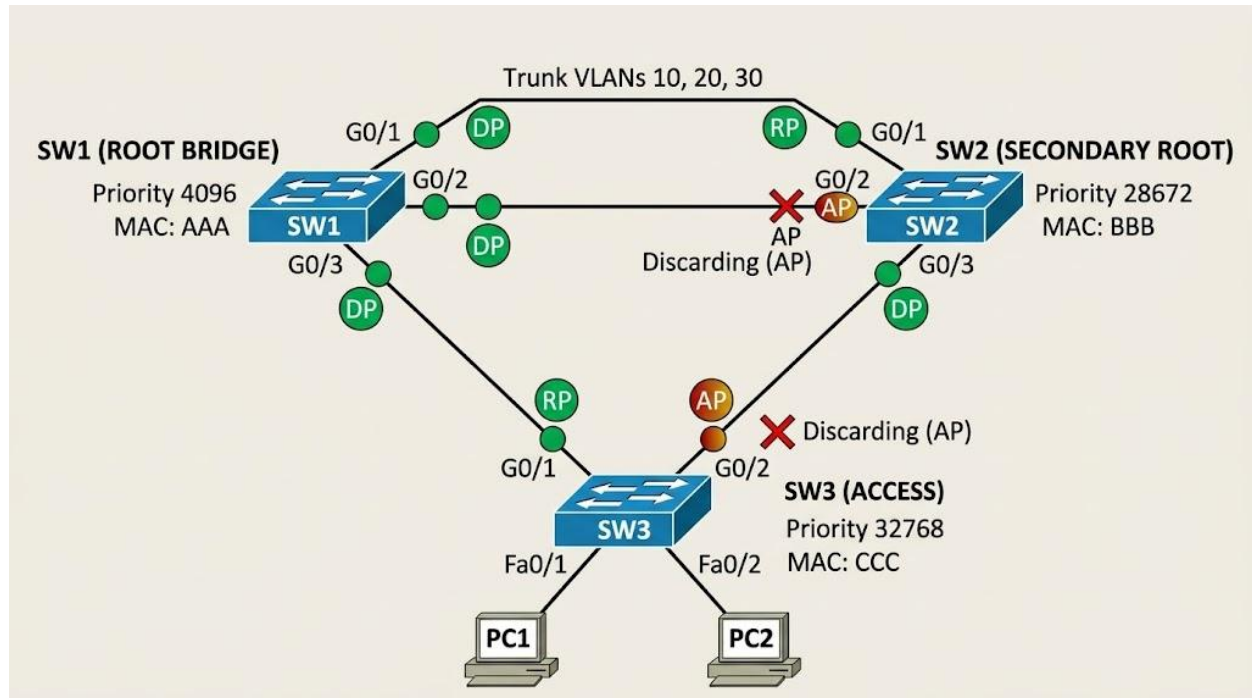
Rapid PVST+ LAB 15

This hands-on lab is designed for CCNA students to build a three-switch Layer-2 topology and practice Rapid PVST+ root bridge selection, root/alternate/designated ports, PortFast, BPDU Guard, Root Guard, and Loop Guard.

Objectives

- Configure Rapid PVST+ using spanning-tree mode rapid-pvst.
- Configure SW1 as the Root Primary and SW2 as the Root Secondary.
- Identify root bridge, root port, designated port, and alternate port.
- Understand Rapid PVST+ port states: Discarding, Learning, and Forwarding.
- Configure PortFast on end-device/access ports.
- Configure BPDU Guard to protect PortFast access ports.
- Configure Root Guard on designated switch-to-switch ports.
- Configure Loop Guard and understand protection against unidirectional/STP failure conditions.
- Verify STP using show commands and perform controlled troubleshooting tests.

Lab Topology



Addressing and VLAN Plan

Device / Port		Purpose	IP / VLAN
SW1 SVI VLAN 99		Management	192.168.99.11/24
SW2 SVI VLAN 99		Management	192.168.99.12/24
SW3 SVI VLAN 99		Management	192.168.99.13/24
PC1		VLAN 10 host	192.168.10.10/24
PC2		VLAN 20 host	192.168.20.10/24
VLAN 10		Students	192.168.10.0/24
VLAN 20		Staff	192.168.20.0/24
VLAN 30		Servers/Lab	192.168.30.0/24
VLAN 99		Management	192.168.99.0/24
Link	SW1	SW2	SW3
SW1–SW2	G0/1	G0/1	—
SW1–SW3	G0/2	—	G0/1
SW2–SW3	—	G0/2	G0/2

End device 1	—	—	Fa0/1
End device 2	—	—	Fa0/2

4. Rapid PVST+ Design

Item	Expected Result
Root Bridge	SW1
Secondary Root	SW2
SW1–SW2	Forwarding/designated on both sides according to VLAN root calculation
SW3 uplink toward SW1	Root Port (Forwarding)
SW3 uplink toward SW2	Alternate Port (normally Discarding)
SW3 access ports	Designated + PortFast + BPDU Guard
SW1/SW2 ports toward SW3	Designated ports with Root Guard
Loop Guard	Applied to appropriate redundant switch-to-switch ports; verify behavior

SSH management access on all three switches using VLAN 99

Switch VLAN 99 Management IP Username Password

SW1	192.168.99.11/24	admin	cisco
SW2	192.168.99.12/24	admin	cisco
SW3	192.168.99.13/24	admin	cisco

SW1– SSH Configuration

enable

configure terminal

hostname SW1

vlan 99

name MANAGEMENT

interface vlan 99

ip address 192.168.99.11 255.255.255.0

no shutdown

username admin privilege 15 secret cisco

ip domain-name iidnetwork.local

crypto key generate rsa modulus 2048

ip ssh version 2

line vty 0 4

login local

transport input ssh

exec-timeout 10 0

end

write memory

SW2 – SSH Configuration

enable

configure terminal

hostname SW2

vlan 99

name MANAGEMENT

interface vlan 99

ip address 192.168.99.12 255.255.255.0

no shutdown

username admin privilege 15 secret cisco

ip domain-name iidnetwork.local

crypto key generate rsa modulus 2048

ip ssh version 2

line vty 0 4

login local

transport input ssh

exec-timeout 10 0

end

write memory

SW3 – SSH Configuration

enable

configure terminal

hostname SW3

vlan 99

name MANAGEMENT

interface vlan 99

ip address 192.168.99.13 255.255.255.0

no shutdown

username admin privilege 15 secret cisco

ip domain-name iidnetwork.local

crypto key generate rsa modulus 2048

ip ssh version 2

line vty 0 4

login local

transport input ssh

exec-timeout 10 0

end

write memory

Basic Configuration on All Switches

SW1:

enable

configure terminal

hostname SW1

no ip domain-lookup

vlan 10

name STUDENTS

vlan 20

name STAFF

vlan 30

name SERVERS

vlan 99

name MANAGEMENT

spanning-tree mode rapid-pvst

interface vlan 99

ip address 192.168.99.11 255.255.255.0

no shutdown

end

write memory

SW2:

```
enable
configure terminal
hostname SW2
no ip domain-lookup
```

```
vlan 10
  name STUDENTS
vlan 20
  name STAFF
vlan 30
  name SERVERS
vlan 99
  name MANAGEMENT
```

```
spanning-tree mode rapid-pvst
```

```
interface vlan 99
ip address 192.168.99.12 255.255.255.0
no shutdown
```

```
end
write memory
```

SW3:

```
enable
configure terminal
hostname SW3
no ip domain-lookup
```

```
vlan 10
  name STUDENTS
vlan 20
  name STAFF
vlan 30
  name SERVERS
vlan 99
  name MANAGEMENT
```

```
spanning-tree mode rapid-pvst
```

```
interface vlan 99
```

```
ip address 192.168.99.13 255.255.255.0
```

```
no shutdown
```

```
end
```

```
write memory
```

Configure the Trunks

SW1:

```
configure terminal
```

```
interface gigabitEthernet 0/1
```

```
description TRUNK_TO_SW2
```

```
switchport mode trunk
```

```
switchport trunk allowed vlan 10,20,30,99
```

```
no shutdown
```

```
interface gigabitEthernet 0/2
```

```
description TRUNK_TO_SW3
```

```
switchport mode trunk
```

```
switchport trunk allowed vlan 10,20,30,99
```

```
no shutdown
```

```
end
```

SW2:

```
configure terminal
```

```
interface gigabitEthernet 0/1
```

```
description TRUNK_TO_SW1
```

```
switchport mode trunk
```

```
switchport trunk allowed vlan 10,20,30,99
```

```
no shutdown
```

```
interface gigabitEthernet 0/2
```

```
description TRUNK_TO_SW3
```

```
switchport mode trunk
switchport trunk allowed vlan 10,20,30,99
no shutdown
end
```

SW3:

```
configure terminal
interface gigabitEthernet 0/1
description TRUNK_TO_SW1
switchport mode trunk
switchport trunk allowed vlan 10,20,30,99
no shutdown
```

```
interface gigabitEthernet 0/2
description TRUNK_TO_SW2
switchport mode trunk
switchport trunk allowed vlan 10,20,30,99
no shutdown
end
```

Configure SW1 as Root Primary and SW2 as Root Secondary

Configure the root bridge per VLAN. Practice with per-VLAN root selection.

SW1 – Root Primary:

```
configure terminal
spanning-tree vlan 10,20,30,99 root primary
end
```

SW2 – Root Secondary:

```
configure terminal
spanning-tree vlan 10,20,30,99 root secondary
end
```

Verification:

show spanning-tree root

show spanning-tree vlan 10

show spanning-tree vlan 20

show spanning-tree vlan 30

show spanning-tree vlan 99

show spanning-tree interface gigabitEthernet 0/1 detail